

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 10: Operations with Integers

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	10 — Operations with Integers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Expert-tier signed-integer reasoning: long mixed chains under strict BODMAS, deeply nested brackets and braces, parity-of-negative-factors arguments, distributive splits, powers where (-2) cubed is not minus 2 cubed, and word problems where every classic slip (subtracting a negative ADDS, two negatives MULTIPLY to a positive, a single hidden factor of 0 collapses the product, and division by the wrong sign) is deliberately baked into the distractors. Do not write on this paper — mark answers on your answer sheet.

1. A counter starts at +5 on a number line. It is moved -8, then -6, then +11, then -4, then +9, then -13. Its final position is:

- (A) -11 (B) +6
(C) -6 (D) -16

2. Compute $(-27) + (+34) + (-19) + (+8) + (-15) + (+6)$.

- (A) -61 (B) +13
(C) -13 (D) +109

3. Which single statement about integer operations is FALSE?

- (A) Subtracting a negative integer is the same as adding its opposite.
(B) The integers are closed under subtraction but not under division.
(C) $a - (b - c)$ need not equal $(a - b) - c$, so subtraction is not associative.
(D) If $a - b$ is positive then $b - a$ must also be positive.

4. Evaluate $(-360) \div (-6) \div (-5) \div (+2) \div (-3)$.

- (A) -2 (B) $+2$
(C) $+6$ (D) -6

5. Evaluate $(-8) + (-5) \times (-6) - (-4) \times (+3) \div (-2)$.

- (A) $+28$ (B) -2
(C) $+16$ (D) $+4$

6. Which expression equals 0 purely by the additive-inverse property, needing no magnitude arithmetic?

- (A) $(-83) \times 0$ (B) $(-83) + 83$
(C) $83 - (-83)$ (D) $(-83) - (-83) - 83$

7. For any integer a , the expression $a \times (-1) \times (-1) \times (-1) \times (-1) \times (-1)$ equals:

- (A) a itself (B) always negative
(C) the opposite of a (D) always 0

8. At 4 a.m. the temperature was -7°C . It then fell 5°C , rose 9°C , fell 8°C , and rose 3°C . The temperature now is:

- (A) -2°C (B) -18°C
(C) $+6^\circ\text{C}$ (D) -8°C

9. Compute $(-2) \times (+3) \times (-1) \times (-4) \times (-5) \times (-1)$.

- (A) -120 (B) $+120$
(C) -15 (D) $+15$

10. A drone at altitude 0 m climbs 18 m, descends 27 m, descends 14 m, climbs 31 m, then descends 9 m. Twice its final altitude is:

- (A) -1 m (B) $+2$ m
(C) -2 m (D) -18 m

11. Which sign-rule conclusion is INCORRECT?

- (A) A negative integer divided by a positive integer gives a negative quotient.
(B) The product of three negative integers is negative.
(C) A negative integer minus a larger-magnitude negative integer can give a positive result.
(D) The product of four negative integers is negative.

12. On a number line you start at $+6$, move 9 left, then 4 right, then 11 left, then 8 right, then 7 left, then 2 right. Where do you land?

- (A) -5 (B) -7
(C) $+7$ (D) -9

13. The product of nine non-zero integers is positive. Which is NOT a possible number of negative factors among them?

- (A) 5
(C) 2
- (B) 0
(D) 8

14. A product of several non-zero integers contains exactly 6 negative factors and the rest positive; one of the positive factors is then changed to its opposite. The new product's sign is:

- (A) positive
(C) unchanged
- (B) zero
(D) negative

15. Evaluate $(-108) \div (-9) + (-7) \times (-4) - (-2) \times (+8) \div (-4)$.

- (A) +36
(C) +28
- (B) +44
(D) +4

16. By how much is (-17) greater than (-63) ?

- (A) 46
(C) 80
- (B) -46
(D) -80

17. Evaluate $(-3)^4 + (-3)^3 - (-3)^2 + (-3)^1$.

- (A) -42
(C) +96
- (B) +42
(D) +60

18. Evaluate $40 - 72 \div (-8) \times 3 + (-54) \div (-6)$.

- (A) +76
(C) +58
- (B) +22
(D) +4

19. Which statement about identities and inverses for integers is TRUE?

- (A) Every non-zero integer has a multiplicative inverse that is also an integer.
(B) 0 is the additive identity and 1 is the multiplicative identity, while -1 is the multiplicative inverse only of itself and of 1's opposite among integers.
(C) 1 is the additive identity because adding 1 leaves a number's sign unchanged.
(D) 0 is the multiplicative identity because any integer times 0 returns the integer.

20. Compute $(-1) \times (-2) \times (-3) \times (-4) \times (-5) \times (-6) \times (-7)$.

- (A) +5040
(C) -5040
- (B) -28
(D) +28

21. A submarine at -720 m rises in equal steps; after 8 steps it reaches -80 m. Each step, and the depth after the 5th step, are:

- (A) +90 m per step; -270 m after 5 steps
(B) -80 m per step; -1120 m after 5 steps
(C) +80 m per step; -400 m after 5 steps
(D) +80 m per step; -320 m after 5 steps

22. For any non-zero integer n , the expression $(n \times (-1)) \times (n \times (-1)) \times (n \times (-1))$ equals:

- (A) $-n^3$
(C) $-3n$
- (B) n^3
(D) always positive

23. Compute $(-14) + (-14) + (-14) + (-14) + (-14) + (-14) + (-14) + (-14) + (-14)$.

- (A) -126 (B) -112
 (C) -140 (D) $+126$

24. Evaluate $(-150) \div [(-3) - (+2)] - [(-4) \times (-3) + (-2)]$.

- (A) $+40$ (B) $+20$
 (C) -10 (D) 0

25. Which property justifies rewriting $(-12) \times 97$ as $(-12) \times 100 + (-12) \times (-3)$?

- (A) the distributive property, using $97 = 100 + (-3)$
 (B) the associative property of multiplication
 (C) the commutative property of addition
 (D) the additive-inverse property

26. Evaluate $(-288) \div 8 \times (-1) \div (-2) \times (-3) + (-5)$.

- (A) -49 (B) $+49$
 (C) $+59$ (D) $+54$

27. In a contest, $+6$ marks are given for each correct answer, -4 for each wrong one, and -1 for each unattempted. Of 25 questions a student gets 14 right, 8 wrong, and leaves 3 unattempted. The total score is:

- (A) $+52$ (B) $+49$
 (C) $+84$ (D) $+45$

28. Which value makes $(-9) \times \underline{\hspace{1cm}} + 13 = -50$ true?

- (A) -7 (B) -4
 (C) 7 (D) 4

29. Evaluate $(-6) - [(-9) - (-4) - (+7)] + [(-2) - (-5)]$.

- (A) -9 (B) $+3$
 (C) -15 (D) $+9$

30. Evaluate $[(-7) - (-15)] \times [(-4) - (+6)] \div [(-2) + (-3)]$.

- (A) -16 (B) $+16$
 (C) $+80$ (D) -80

31. On a number line, order from LEAST to GREATEST: $(-4) \times (-3)$, $(-24) \div 4$, $(-2) \times (-2) \times (-2)$, $(-18) \div (-6)$.

- (A) $(-2) \times (-2) \times (-2)$, $(-24) \div 4$, $(-18) \div (-6)$, $(-4) \times (-3)$
 (B) $(-24) \div 4$, $(-2) \times (-2) \times (-2)$, $(-18) \div (-6)$, $(-4) \times (-3)$
 (C) $(-4) \times (-3)$, $(-18) \div (-6)$, $(-24) \div 4$, $(-2) \times (-2) \times (-2)$
 (D) $(-2) \times (-2) \times (-2)$, $(-18) \div (-6)$, $(-24) \div 4$, $(-4) \times (-3)$

32. The sum of an integer and its opposite is multiplied by 47, and then 19 is subtracted. The result is:

- (A) 0 (B) -19
 (C) $+19$ (D) $+47$

33. On a number line you start at -4 , then your position is multiplied by 3 (keeping its sign), then you move $+20$, then your position is multiplied by (-2) . Where do you end up?

- (A) $+16$ (B) -40
(C) -16 (D) $+8$

34. Evaluate $(-2)^3 \times (-3)^2 - (-4)^2 \div (-2) + (-1)^7$.

- (A) -81 (B) -65
(C) -63 (D) -49

35. Which sequence of three integer operations on the number 10 , applied in order, returns to 10 for EVERY starting integer when applied to that integer instead?

- (A) add (-10) , then multiply by (-1) , then add 10
(B) multiply by 0 , then add 10 , then multiply by 1
(C) subtract 5 , then multiply by (-2) , then add 0
(D) multiply by (-1) , then multiply by (-1) , then add 0

36. The product $(-1) \times (-2) \times (-3) \times \dots \times (-12) \times 0$ (all integers from -1 down to -12 , times 0) is:

- (A) positive (B) negative
(C) zero (D) -78

37. Evaluate $(-5)^3 \div (-5)^2 \times (-2) + (-3)^2 \times (-1)^3$.

- (A) -1 (B) $+1$
(C) -19 (D) $+19$

38. A delivery rider's account changes by $-\text{₹}240$, $+\text{₹}180$, $-\text{₹}95$, $+\text{₹}310$, $-\text{₹}150$ during a shift. The net change, then doubled, equals:

- (A) $+\text{₹}5$ (B) $+\text{₹}10$
(C) $-\text{₹}10$ (D) $+\text{₹}190$

39. If p, q, r, s are all negative integers, then $p \times q \times r \times s \times (p + q + r + s)$ is:

- (A) positive (B) negative
(C) zero (D) cannot be determined

40. Evaluate $(19 - 34) \times (-4) + (-96) \div (-8) - (-3) \times (+5)$.

- (A) $+57$ (B) $+45$
(C) $+27$ (D) $+87$

41. The product $(-1) \times (-3) \times (-5) \times (-7) \times (-9) \times (-11)$ (the first six odd negatives) is:

- (A) negative (B) zero
(C) positive (D) -36

42. Which value makes $(-156) \div \underline{\hspace{1cm}} + 7 = -6$ true?

- (A) 12 (B) -12
(C) $+24$ (D) -13

43. The integers are closed under three of the four basic operations. Which statement correctly captures closure?

- (A) Dividing, multiplying, or adding any two integers always gives an integer, but subtracting may not.
- (B) All four operations on any two integers always give an integer.
- (C) Only addition and multiplication of integers always give an integer.
- (D) Adding, subtracting, or multiplying any two integers always gives an integer, but dividing two integers may not.

44. Evaluate $(-3) \times [(-2)^3 + (-4) \times (-3) - (-5)] - (-2)^4$.

- (A) -11
- (B) -43
- (C) +43
- (D) -5

45. Two integers have a sum of -9 and a product of +20. The integers are:

- (A) 5 and 4
- (B) -10 and 1
- (C) -20 and -1
- (D) -5 and -4

46. Evaluate $(-23) - (-38) - (+17) - (-9) - (+12) - (-6)$.

- (A) -1
- (B) -39
- (C) +1
- (D) +39

47. Evaluate $(-196) \div (-7) \div (-2) - (-5) \times (-3) + (-4)^2$.

- (A) -13
- (B) +1
- (C) -29
- (D) -43

48. Which expression equals the SMALLEST (most negative) integer?

- (A) $(-2) \times (-2) \times (-2) \times (-2) \times (-2)$
- (B) $(-3) \times (-3) \times (-3)$
- (C) $(-5) \times (-5)$
- (D) $(-9) - (-4) - (+6)$

49. A bank account at ₹0 records: debit ₹450, credit ₹275, debit ₹130, credit ₹600, debit ₹315. The closing balance is:

- (A) +₹20
- (B) -₹650
- (C) -₹20
- (D) +₹875

50. Evaluate $(-13) - 7 \times (-5) + (-72) \div (-9) \times 2 - 6$.

- (A) +16
- (B) +48
- (C) +20
- (D) +32

51. A lift starts at floor +6 (basements negative). It goes down 11, up 5, down 8, up 3, down 7, up 4. Three floors below where it ends is floor:

- (A) -8
- (B) -5
- (C) -14
- (D) -11

52. Evaluate $(-18) \times (-5) - (-18) \times (+3) + (-18) \times (-2) - (-18) \times (+4)$, then divide the result by (-18) .

- (A) -14 (B) $+14$
(C) -4 (D) $+4$

53. Evaluate $-6^2 + (-6)^2 - (-2)^3 + (-1)^8$.

- (A) -9 (B) $+81$
(C) $+9$ (D) -7

54. Evaluate $[(-5) \times (-4) - 8] \div [(-2) + (-1)] \times (-3) - (-4)$.

- (A) $+8$ (B) -8
(C) $+12$ (D) $+16$

55. Two integers have a sum of $+2$ and a product of -48 . The integers are:

- (A) 8 and -6 (B) -8 and 6
(C) 12 and -4 (D) -12 and 4

56. An odd number of negative integers are multiplied together with exactly one factor equal to 0 and the rest positive. The product is:

- (A) negative (B) positive
(C) depends on the count of negatives (D) zero

57. A weather station logs hourly changes of $-3, -2, +6, -4, -1, +2, -5, +3$ °C, starting from 5 °C. The final temperature, then lowered by a further 4 °C, is:

- (A) $+1$ °C (B) -7 °C
(C) $+5$ °C (D) -3 °C

58. Evaluate $(-10) - \{(-6) - [(-8) + (-3) - (-5)]\} + (-2)$.

- (A) -18 (B) -6
(C) -12 (D) $+6$

59. The product of an EVEN number of negative integers, multiplied by an ODD number of further negative integers, with all factors non-zero, is always:

- (A) positive (B) zero
(C) negative (D) even

60. Evaluate $[(-9) \times (-2) - 6] \div [(-4) + (+1)] \times (-5) \div (+2) - (-7)$.

- (A) $+17$ (B) $+3$
(C) -17 (D) $+27$

Solutions — Mathematics (Class 7), Chapter 10: Operations with Integers — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	D	31	A	41	C	51	D
2	C	12	B	22	A	32	B	42	A	52	A
3	D	13	A	23	A	33	C	43	D	53	C
4	B	14	D	24	B	34	B	44	B	54	D
5	C	15	A	25	A	35	D	45	D	55	A
6	B	16	A	26	B	36	C	46	C	56	D
7	C	17	B	27	B	37	B	47	A	57	D
8	D	18	A	28	C	38	B	48	A	58	C
9	A	19	B	29	D	39	B	49	C	59	C
10	C	20	C	30	B	40	D	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Sum the moves with the start: $5 + (-8) + (-6) + 11 + (-4) + 9 + (-13)$. Group the negatives: $-8 -6 -4 -13 = -31$; group the positives with the start: $5 + 11 + 9 = 25$. So $25 + (-31) = -6$. A common slip drops the start value or mishandles a sign, giving -11 or -16 ; another reads the answer with the wrong sign to get $+6$.

2. (C) Positives: $34 + 8 + 6 = 48$. Negatives: $-27 -19 -15 = -61$. Total: $48 + (-61) = -13$. Adding all magnitudes ignoring signs gives 109; combining only the negatives gives -61 ; reporting the right size with the wrong sign gives $+13$.

3. (D) Since $b - a$ is the opposite of $a - b$, if one is positive the other is negative; so the claim that both are positive is false. The other three are all true: subtracting a negative adds, integers are closed under subtraction but division can leave the integers, and subtraction is not associative.

4. (B) Work left to right: $(-360) \div (-6) = 60$; $60 \div (-5) = -12$; $(-12) \div 2 = -6$; $(-6) \div (-3) = 2$. There are three negative divisors plus the negative dividend — four negatives in all

gives a positive result. Stopping at -6 or mismanaging the parity of signs yields -2 , $+6$, or -6 .

5. (C) Do $\times$ and $\div$ first, left to right. $(-5) \times (-6) = 30$. For the last term, $(-4) \times (+3) = -12$, then $-12 \div (-2) = 6$; it is subtracted: -6 . So $(-8) + 30 - 6 = 16$. Treating the subtracted 6 as $+6$ gives 28; mishandling the first product's sign gives -2 .

6. (B) An integer plus its own opposite is 0 by the additive-inverse property: $(-83) + 83 = 0$. Multiplying by 0 gives 0 but by the zero-product rule, not the inverse property. The third equals 166, and the fourth equals -83 , neither being 0.

7. (C) There are five factors of (-1) ; an odd count of negatives makes the product negative, so the (-1) factors multiply to -1 , giving $a \times (-1) = -a$, the opposite of a . With an even count it would be a itself, but five is odd. It is not always negative (depends on a 's sign) nor 0.

8. (D) Start -7 , then -5 , $+9$, -8 , $+3$. Sum the changes: $-5 + 9 - 8 + 3 = -1$. So $-7 + (-1) = -8$ °C. Forgetting the start and summing only changes gives -1 ; misreading a fall as a rise can give -2 or $+6$.

9. (A) Magnitudes: $2 \times 3 \times 1 \times 4 \times 5 \times 1 = 120$. Count the negative factors: -2 , -1 , -4 , -5 , -1 — that is five negatives (the $+3$ is positive). Five is odd, so the product is negative: -120 . An even miscount gives $+120$; adding magnitudes mistakenly gives ± 15 .

10. (C) Changes: $+18 - 27 - 14 + 31 - 9$. Climbs: $18 + 31 = 49$. Descents: $27 + 14 + 9 = 50$. Net = $49 - 50 = -1$ m. Twice this is -2 m. Forgetting to double gives -1 ; sign error on the doubling gives $+2$.

11. (D) An even number of negative factors gives a positive product, so four negatives multiply to a positive — the claim that it is negative is wrong. The others hold: opposite signs divide to negative, three negatives (odd) multiply to negative, and e.g. $(-2) - (-9) = +7$.

12. (B) Net = $6 - 9 + 4 - 11 + 8 - 7 + 2$. Rights (with start): $6 + 4 + 8 + 2 = 20$. Lefts: $9 + 11 + 7 = 27$. So $20 - 27 = -7$. Dropping a step or swapping a direction gives -5 , $+7$, or -9 .

13. (A) A positive product needs an EVEN number of negative factors. Among 0 through 9, the even options 0, 2, and 8 are all possible; 5 is odd and would force a negative product, so it is impossible.

14. (D) Originally 6 negatives (even) $\rightarrow$ positive product. Turning one positive factor negative raises the negative count to 7 (odd), so the product becomes negative. It is not zero (no factor is 0), and the sign does change.

15. (A) First term: $(-108) \div (-9) = 12$. Second: $(-7) \times (-4) = 28$. Third: $(-2) \times 8 = -16$, then $-16 \div (-4) = 4$, subtracted $\rightarrow -4$. Total: $12 + 28 - 4 = 36$. Adding the last term instead of subtracting gives 44; sign slips give 28 or 4.

16. (A) "How much greater" means $(-17) - (-63) = -17 + 63 = 46$. Adding the magnitudes gives 80; reversing the subtraction gives -46 .

17. (B) $(-3)^4 = 81$ (even power, positive), $(-3)^3 = -27$, $(-3)^2 = 9$, $(-3)^1 = -3$. So $81 + (-27) - 9 + (-3) = 81 - 27 - 9 - 3 = 42$. Treating odd powers as positive, or $-(-3)^2$ as $+9$, gives the larger wrong values.

18. (A) Left to right for $\times$ and $\div$: $72 \div (-8) = -9$, then $-9 \times 3 = -27$, and it is subtracted: $40 - (-27) = 40 + 27 = 67$. Last term: $(-54) \div (-6) = 9$. Total $67 + 9 = 76$. Reading $40 - (-27)$ as $40 - 27$ gives 22; other sign slips give 58 or 4.

19. (B) $a + 0 = a$ and $a \times 1 = a$, so 0 and 1 are the additive and multiplicative identities; the only integers that are their own reciprocals are 1 and -1 . Most integers have no integer reciprocal (e.g. 3's reciprocal is $1/3$), 1 is not the additive identity, and any integer $\times 0 = 0$, not the integer.

20. (C) Magnitude is $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 = 5040$. There are seven negative factors; seven is odd, so the product is negative: -5040 . An even miscount gives $+5040$; summing instead of multiplying gives ± 28 .

21. (D) Total rise = $(-80) - (-720) = 640$ m over 8 steps, so each step is $640 \div 8 = +80$ m. After 5 steps: $-720 + 5 \times 80 = -720 + 400 = -320$ m. Using $640 \div 8$ wrongly as 90, or stopping at -400 (the rise, not the depth), are the traps.

22. (A) Each bracket is $-n$, so the product is $(-n) \times (-n) \times (-n) = (-1)^3 \times n^3 = -n^3$ (three negatives $\rightarrow$ negative). It is not n^3 nor $-3n$, and its sign follows n^3 , so it is not always positive.

23. (A) Nine copies of -14 : $9 \times (-14) = -126$. Miscounting as 8 or 10 gives -112 or -140 ; sign error gives $+126$.

24. (B) First bracket: $(-3) - 2 = -5$, so $(-150) \div (-5) = 30$. Second bracket: $(-4) \times (-3) = 12$, $12 + (-2) = 10$, subtracted $\rightarrow -10$. So $30 - 10 = 20$. Forgetting to subtract the second bracket gives 30 then 40; sign slips give -10 or 0.

25. (A) Writing $97 = 100 + (-3)$ and multiplying each part by (-12) is distribution of multiplication over addition. The result is $-1200 + 36 = -1164$. The associative, commutative, and inverse properties describe different rearrangements, not this expansion.

26. (B) Work $\times$ and $\div$ strictly left to right. $(-288) \div 8 = -36$; $-36 \times (-1) = 36$; $36 \div (-2) = -18$; $-18 \times (-3) = 54$. Then $54 + (-5) = 49$. Stopping before the $+(-5)$ gives 54; a single sign slip in the chain flips it to -49 ; misreading $+(-5)$ as $-(-5)$ gives 59.

27. (B) Correct: $14 \times 6 = 84$. Wrong: $8 \times (-4) = -32$. Unattempted: $3 \times (-1) = -3$. Total: $84 - 32 - 3 = 49$. Ignoring the unattempted penalty gives 52; counting only the positive marks gives 84; other slips give 45.

28. (C) $(-9) \times x + 13 = -50 \rightarrow (-9)x = -63 \rightarrow x = (-63) \div (-9) = 7$. Forgetting to move the 13 gives x from $(-9)x = -50$; sign errors in the final division give -7 .

29. (D) First bracket: $(-9) - (-4) - 7 = -9 + 4 - 7 = -12$. Second bracket: $(-2) - (-5) = -2 + 5 = 3$. So $(-6) - (-12) + 3 = -6 + 12 + 3 = 9$. Reading $-(-12)$ as -12 gives the negative answers.

30. (B) Brackets: $(-7) - (-15) = 8$; $(-4) - 6 = -10$; $(-2) + (-3) = -5$. Then $8 \times (-10) = -80$; $(-80) \div (-5) = 16$. Stopping at -80 (forgetting the division) or sign slips give the other options.

31. (A) Evaluate: $(-4) \times (-3) = 12$; $(-24) \div 4 = -6$; $(-2) \times (-2) \times (-2) = -8$; $(-18) \div (-6) = 3$. Sorted least $\rightarrow$ greatest: $-8, -6, 3, 12$, i.e. the cube, then -6 , then 3 , then 12 . Mixing up -8 and -6 , or descending order, gives the distractors.

32. (B) An integer plus its opposite is 0; $0 \times 47 = 0$; then $0 - 19 = -19$. Forgetting the final subtraction gives 0; sign error gives $+19$.

33. (C) Start -4 ; $\times 3 \rightarrow -12$; $+20 \rightarrow 8$; $\times(-2) \rightarrow -16$. Forgetting the $\times(-2)$ sign gives $+16$; multiplying -12 by -2 before adding gives wrong paths to ± 40 or $+8$.

34. (B) $(-2)^3 = -8$, $(-3)^2 = 9$, so $-8 \times 9 = -72$. $(-4)^2 = 16$, $16 \div (-2) = -8$, subtracted $\rightarrow -(-8) = +8$. $(-1)^7 = -1$. Total: $-72 + 8 + (-1) = -65$. Reading $-(-4)^2 \div (-2)$ as -8 (not $+8$) gives -81 ; treating $(-1)^7$ as $+1$ gives -63 .

35. (D) For ANY starting integer n : $\times(-1)$ then $\times(-1)$ gives n (two negatives cancel), and $+0$ leaves it unchanged — so it returns n every time. The others depend on the specific value (e.g. $\times 0$ destroys n) and do not return every integer to itself.

36. (C) Any product containing a factor of 0 is 0, regardless of how many negative factors there are. The hidden 0 collapses everything; sign-parity reasoning about the twelve negatives is irrelevant. It is not -78 (that is the sum $-1-2-\dots-12$).

37. (B) $(-5)^3 = -125$, $(-5)^2 = 25$, so $-125 \div 25 = -5$; then $\times(-2) = 10$. For the second term, $(-3)^2 = 9$ and $(-1)^3 = -1$, so $9 \times (-1) = -9$. Total: $10 + (-9) = +1$. A sign slip on the first term gives -1 ; treating $(-1)^3$ as $+1$ gives $+19$; treating $(-5)^3 \div (-5)^2$ as $+5$ gives -19 .

38. (B) Credits: $180 + 310 = 490$. Debits: $240 + 95 + 150 = 485$. Net = $490 - 485 = +5$. Doubled = $+10$. Forgetting to double gives $+5$; sign error gives -10 ; adding all magnitudes gives wrong values.

39. (B) $p \times q \times r \times s$ is a product of four negatives (even) $\rightarrow$ positive. The sum $p+q+r+s$ of four negatives is negative. Positive $\times$ negative = negative. It is never zero (no factor is 0), and the sign is always determined.

40. (D) $(19 - 34) = -15$, $\times (-4) = 60$. $(-96) \div (-8) = 12$. $(-3) \times 5 = -15$, subtracted $\rightarrow +15$. Total: $60 + 12 + 15 = 87$. Subtracting 15 instead of adding gives 57; dropping a term gives 45 or 27.

41. (C) Six negative factors — six is even — so the product is positive (it equals $+10395$). An odd miscount would say negative; there is no zero factor; -36 would be the sum, not the product.

42. (A) $(-156) \div x + 7 = -6 \rightarrow (-156) \div x = -13 \rightarrow x = (-156) \div (-13) = 12$. Sign error in the final division gives -12 ; ignoring the $+7$ gives a different x .

43. (D) Integers are closed under $+$, $-$, and $\times$, but not $\div$ (e.g. $7 \div 2$ is not an integer). Subtraction IS closed, so the first distractor is wrong; not all four are closed; and subtraction belongs in the closed set too, so 'only $+$ and $\times$ ' is wrong.

44. (B) Inside: $(-2)^3 = -8$; $(-4) \times (-3) = 12$; $-(-5) = +5$; sum = $-8 + 12 + 5 = 9$. Then $(-3) \times 9 = -27$. $(-2)^4 = 16$, subtracted $\rightarrow -16$. Total: $-27 - 16 = -43$. Treating $(-2)^4$ as -16 (so $-27 - (-16)$) gives -11 ; sign error gives $+43$.

45. (D) A positive product with a negative sum means both integers are negative. We need two negatives multiplying to 20 and summing to -9 : -5 and -4 give product 20 and sum -9 . The pair 5 and 4 sums to $+9$; -10 and 1 give product -10 ; -20 and -1 sum to -21 .

46. (C) Rewrite as adds of opposites: $-23 + 38 - 17 + 9 - 12 + 6$. Positives: $38 + 9 + 6 = 53$. Negatives: $-23 - 17 - 12 = -52$. Total: $53 - 52 = +1$. A single sign slip flips it to -1 or shifts the magnitude to ± 39 .

47. (A) $(-196) \div (-7) = 28$; $28 \div (-2) = -14$. $(-5) \times (-3) = 15$, subtracted $\rightarrow -15$. $(-4)^2 = +16$. Total: $-14 - 15 + 16 = -13$. Mishandling the chained division sign gives $+1$; treating $(-4)^2$ as -16 gives the lower values.

48. (A) Evaluate: $(-2)^5 = -32$; $(-3)^3 = -27$; $(-5)^2 = +25$; $(-9) - (-4) - 6 = -9 + 4 - 6 = -11$. The most negative is -32 . Treating $(-2)^5$ as positive, or stopping at $(-3)^3$, would mislead.

49. (C) Credits: $275 + 600 = 875$. Debits: $450 + 130 + 315 = 895$. Balance = $875 - 895 = -\text{₹}20$. Sign error gives $+20$; summing only one side gives -650 or $+875$.

50. (D) $7 \times (-5) = -35$, subtracted $\rightarrow +35$: $(-13) + 35 = 22$. Next: $(-72) \div (-9) = 8$, $\times 2 = 16$, added $\rightarrow 38$. Then $-6 \rightarrow 32$. Forgetting to $\times 2$ the division term gives 16-less paths; sign slips give 48 or 20.

51. (D) Net = $6 - 11 + 5 - 8 + 3 - 7 + 4$. Ups (with start): $6 + 5 + 3 + 4 = 18$. Downs: $11 + 8 + 7 = 26$. End = $18 - 26 = -8$. Three floors below: $-8 - 3 = -11$. Forgetting the extra 3 gives -8 ; sign error on it gives -5 .

52. (A) Factor out (-18) using the distributive property: $(-18) \times [(-5) - (+3) + (-2) - (+4)]$. The bracket is $-5 - 3 - 2 - 4 = -14$, so the result is $(-18) \times (-14) = 252$. Dividing by (-18) : $252 \div (-18) = -14$. (Equivalently, dividing the whole product by (-18) just returns the bracket, -14 .) A sign slip gives $+14$; mis-summing the bracket gives ± 4 .

53. (C) -6^2 means $-(36) = -36$; $(-6)^2 = +36$; $(-2)^3 = -8$, subtracted $\rightarrow +8$; $(-1)^8 = +1$. Total: $-36 + 36 + 8 + 1 = 9$. Treating -6^2 as $+36$ gives 81; sign error on $(-2)^3$ gives -7 .

54. (D) Numerator bracket: $(-5) \times (-4) = 20$, $20 - 8 = 12$. Denominator bracket: $(-2) + (-1) = -3$. So $12 \div (-3) = -4$, $\times (-3) = 12$. Then $-(-4) = +4$: $12 + 4 = 16$. Forgetting the final $+4$ gives 12; sign slips give ± 8 .

55. (A) A negative product means opposite signs; a positive sum means the positive number has the larger magnitude. We need them to multiply to -48 and add to $+2$: $8 + (-6) = 2$ and $8 \times (-6) = -48$. The pair -8 and 6 sums to -2 ; 12 and -4 sum to $+8$; -12 and 4 sum to -8 .

56. (D) Any product with a factor of 0 equals 0, no matter the parity of the negative factors. The hidden 0 overrides the sign reasoning, so it is neither negative nor positive and does not depend on the count.

57. (D) Changes sum: $-3 - 2 + 6 - 4 - 1 + 2 - 5 + 3 = -4$. Final after start: $5 + (-4) = 1^\circ\text{C}$. Lowered by 4 more: $1 - 4 = -3^\circ\text{C}$. Forgetting the extra 4 gives $+1$; sign error on it gives $+5$.

58. (C) Innermost: $(-8) + (-3) - (-5) = -8 - 3 + 5 = -6$. Next: $(-6) - (-6) = 0$. Then $(-10) - \{0\} + (-2) = -10 - 0 - 2 = -12$. Mishandling the double negative inside gives -18 or -6 .

59. (C) Even negatives + odd negatives = odd total number of negative factors, and an odd count gives a negative product. No factor is 0, so it is not zero; 'even' describes a count, not a sign.

60. (A) Numerator bracket: $(-9) \times (-2) = 18$, $18 - 6 = 12$. Denominator bracket: $(-4) + 1 = -3$. So $12 \div (-3) = -4$; $\times (-5) = 20$; $\div 2 = 10$. Then $-(-7) = +7$: $10 + 7 = 17$. Forgetting the final $+7$ gives 10-based slips; sign error gives -17 .